

PSG Institute of Technology and Applied Research, Coimbatore - 641 062

Internal Assessment Test - 1
CCS334 – Recommender Systems
 III- B.Tech (AI&DS) Fifth Semester

Date: 26.09.2025 FN

Duration: 2 hours

No. of Pages : 3

Maximum: 65 Marks

Answer ALL questions**PART A – (10 × 2 = 20 marks)**

- 1 A fashion e-commerce website displays a “Winter Collection” featuring scarves, jackets, and boots to all users uniformly during the winter season. Identify the type of recommender system approach used here: personalized or non-personalized? Mention one advantage of this approach during seasonal promotions. (K3) CO1
- 2 Given three users have rated a product as 3, 4, and 5 respectively, and the global average rating is 3.5, calculate the damped mean rating for the product using a damping factor of 5. (K3) CO1
- 3 Consider $\lambda=0.1$. If a user gave a rating 5 for an item 3 days ago and a rating 3 for the same item 10 days ago, calculate the time weighted average rating. (K3) CO1
- 4 Given two item vectors representing features in a recommender system: Item A: (3, 4, 5), Item B: (1, 7, 9) Calculate the Euclidean distance between Item A and Item B. (K3) CO1
- 5 Identify the similarity measure that focusses on direction rather than magnitude in recommender system. what is the advantage of prioritizing direction rather than magnitude. (K2) CO1
- 6 A user has liked the following three movies with these item profile vectors (features represent genres; values are normalized frequencies): (K3) CO2

Feature	Movie 1	Movie 2	Movie 3
Action	0.6	0.2	0.3
Comedy	0.2	0.7	0.3
Drama	0.2	0.1	0.4

Construct the user's profile vector.

- 7 Outline the data mining algorithms used in recommender system. (K1) CO2
- 8 Consider a content-based recommender that classifies movies as "Liked" or "Not Liked" by a user based on two binary features: "Action" (1 if action genre, 0 otherwise) and "Comedy" (1 if comedy genre, 0 otherwise). (K3) CO2

Action	Comedy	Class (Liked?)
1	0	Yes
1	1	Yes
0	1	No
0	0	No

Calculate the probability that a new movie with features Action=1 and Comedy=0 will be "Liked" or "Not Liked" by the user using Naive Bayes classification.

- 9 Describe hybrid recommender system. (K1) CO3
- 10 Differentiate between user-based and item-based collaborative filtering. (K2) CO3

PART B - (2× 15 = 30 marks)

- 11 a. A leading e-commerce website wants to enhance its user experience by providing personalized recommendations. Users can create accounts, browse products in various categories (such as electronics, clothing, and books), and rate or purchase items. The platform also has many first-time visitors who have not provided any personal data but frequently view trending products. Analyze this scenario and identify all relevant types (classes) in the taxonomy of recommender systems that could be effectively applied. For each type identified, justify its suitability for a specific user group or business objective within the scenario. Compare and contrast traditional and non-personalized recommender systems with clear examples. (15) (K4) CO1

(OR)

- 11 b. Consider the following 3x2 user-item matrix A: (K4) CO1
- $$A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \\ 0 & 5 \end{bmatrix}$$
- Calculate the singular and eigen value matrix of A using SVD. Using the largest singular value, explain how it would be used to reduce dimensionality in a recommender system. (15)

- 12 a. Design a content-based recommendation system for a popular food delivery app. The app offers a variety of food items from multiple restaurants with features such as cuisine type, ingredients, dietary preferences, price range, and user ratings. Provide a detailed analysis of how you would implement the content-based recommender system using the standard architecture. Specifically, discuss each of the following architectural components in the context of this food delivery app: (K4) CO2
- Content Analyzer: How would you represent food item profiles and extract relevant features?
- Profile Learner: What methods would be used to learn and update user profiles based on their order history, preferences, and feedback?
- Filtering Component: How would similarity-based retrieval work to recommend food items matching user tastes?
- Classification Algorithms: How could classification be employed to predict user preferences or categorize food items?
- Explain the role of each component, how they interconnect, and highlight challenges unique to the food delivery domain. (15)

(OR)

- 12 b. Consider a food delivery app with the following five dishes described by the frequency of keywords in their descriptions: (K4) CO2

Term	Dish 1	Dish 2	Dish 3	Dish 4	Dish 5
spicy	3	0	1	0	2
chicken	1	2	0	1	1
curry	1	1	3	0	0
rice	0	1	1	2	1

vegetarian	0	0	0	3	0
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Build a content based recommender system. Construct the TF-IDF vector for all the dishes. Compute the cosine similarity between Dish 1 and each of the other dishes. Based on these similarities, recommend the top 2 dishes to a user who liked Dish 1. Discuss the advantages and limitations of using TF-IDF and cosine similarity in this content-based recommendation scenario. **(15)**

PART C - (1× 15 = 15 marks)

- 13 a Consider an online bookstore that wants to develop a new recommender system to help customers discover books based on their reading preferences and past purchases. Describe a systematic approach to collaborative filtering based recommender system for the bookstore. **(15)**

(K3)

CO3

(OR)

- 13 b.

	I1	I2	I3	I4	I5	I6
U1	5	3	4	4	?	2
U2	3	1	2	?	3	3
U3	4	3	4	3	5	?
U4	3	3	?	5	3	2
U5	1	5	4	4	2	?

(K4)

CO3

Suppose the above matrix represents ratings (on a scale of 1-5) by 5 users (U1-U5) for 6 items (I1-I6). The value "?" indicates a missing rating. For user U1, calculate the Pearson correlation and cosine similarity between U1 and each of the other users, considering only the items they both rated. Using user-based collaborative filtering, predict the missing rating that user U1 would most likely give to item I5, showing the calculation steps for weighted Pearson (Assume a weighted sum over the top 2 most similar users.). Briefly discuss which similarity measure, Pearson or cosine, better handles differences in user rating scale and why. **(15)**
